

RAMAKRISHNA K

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OBJECTIVE

M.Tech student specializing in VLSI and Embedded Systems, with a B.Tech in Electronics and Communication Engineering. Experienced in embedded hardware, PCB design, robotics, IoT, and power electronics — developing electronic systems from concept to working prototype, with a strong interest in hardware design and VLSI technologies.

EDUCATION

M.Tech — VLSI and Embedded Systems

Aug 2026 – Present

Pondicherry Technological University (PTU), Puducherry

B.Tech — Electronics and Communication Engineering

Aug 2022 – May 2026

Acharya College of Engineering and Technology, Puducherry

CGPA: 8.46 / 10 — First Class

Higher Secondary Certificate (HSC)

May 2022

Sri Ramakrishna Matric Higher Secondary School — CGPA: 7.5 / 10

Secondary School Leaving Certificate (SSLC)

Apr 2020

CGPA: 7.5 / 10

TECHNICAL SKILLS

Microcontrollers	Arduino, ESP32, ESP8266, STM32, Raspberry Pi
PCB Design	EasyEDA, KiCad
Simulation Tools	Proteus, Multisim
Embedded Software	Arduino IDE, Keil µVision
Programming	C, Embedded C
Hardware Design	Schematic Design, Circuit Debugging, Sensor Interfacing, Prototyping
Power Electronics	DC–DC Converters, MOSFET Switching, Motor Drivers, Battery Systems
Testing & Tools	Oscilloscope, Digital Multimeter, Soldering, Troubleshooting

PROJECTS

IRIS — Intelligent Robot with Integrated Sensors

- Developing an autonomous indoor college robot for human interaction and lightweight document delivery.
- Integrated Intel i7 onboard computer, Arduino Mega, webcam, ultrasonic sensors, LCD, speaker, and servo-actuated head and arms.
- Designed a custom high-current motor driver using IRF3205 MOSFETs and IR2104 gate drivers for differential-drive motors.
- Implemented sensor interfacing, robotic motion control, human interaction, and AI-assisted functionality.

Fire-Fighting Robot using Arduino

- Built a compact robot that detects fire using IR sensors and activates a water pump through Arduino-based control.

Smart Home Automation using ESP32

- Developed a Wi-Fi-based appliance control system using ESP32 with mobile-based manual and automated switching.

DIY Digital Multimeter

- Designed an Arduino-based digital multimeter with GLCD interface for voltage, resistance, and continuity measurements.

DC Home — 12V Backup Power System

- Developed a 12V rechargeable backup system for essential home loads including LED lighting and DC fans with automatic/manual power switching.

INTERNSHIPS & TRAINING

- R&D Intern — Chozha Win Technology, Puducherry

Apr 2026 – Present

 - Working in hardware design and development for defence-oriented electronic systems.
 - Involved in practical circuit development, testing, debugging, and hardware implementation.
- Industrial Exposure — Rane Madras Pvt. Ltd.

Apr 2025

 - Gained hands-on exposure to PLC-based industrial automation and automotive manufacturing processes.
- IoT Internship

Dec 2024 – Mar 2025

 - Worked on sensor interfacing, data logging, and cloud connectivity for IoT-based applications.
- Embedded/IoT Training — MSME, Kalapet, Puducherry

Aug 2025

 - Received hands-on training in microcontrollers, IoT integration, and embedded system development.
- PLC Training — MSME, Kalapet, Puducherry

Sep 2025

 - Learned PLC programming, industrial automation concepts, and practical control-system applications.

WORKSHOPS

- Conducted a **Basics of Electronics Workshop** — covering component testing, circuit design, and breadboarding.
- Conducted a **PCB Design and Fabrication Workshop** at ACET — practical training in single-layer PCB design and etching.

CERTIFICATIONS

- NPTEL — Introduction to IoT: 83% — Elite + Silver Medal
- NPTEL — Introduction to Industrial IoT (Industry 4.0): 86% — Elite + Silver Medal

HACKATHONS

- Atom Quest 2025 Hackathon — Atomberg: Selected for the Semi-Final Round among 18,000+ participants nationwide.

ACHIEVEMENTS

- Built and showcased the IRIS robotic platform in technical exhibitions and academic demonstrations.
- Designed and fabricated a custom motor driver capable of handling ~10A continuous current for IRIS.
- Built a 12.6V 36Ah lithium-ion battery pack by repurposing recovered laptop cells for the IRIS robot.
- Designed and maintained a 12V DC home backup power system for essential loads.

LANGUAGES

- English
- Tamil